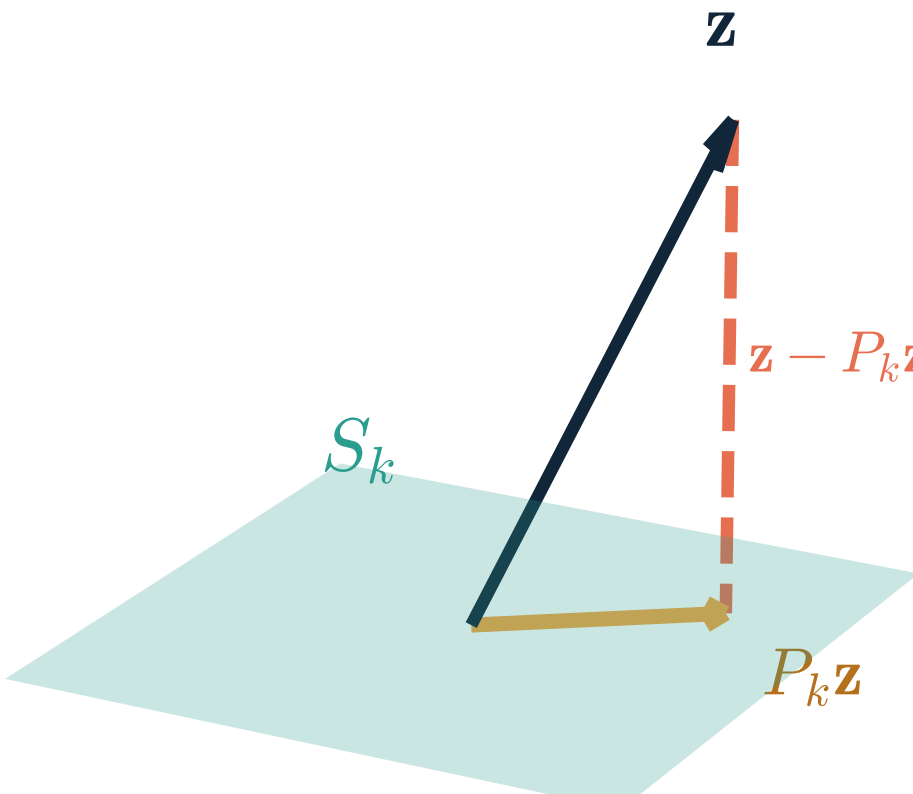
$\mathbf{z}$
$\mathbf{z} - P_k\mathbf{z}$
S_k
$P_k\mathbf{z}$